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Paleomagnetic and isotopic dating of thrust-belt deformation along the eastern edge of the Helena salient, northern Crazy Mountains Basin, Montana

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ABSTRACT

Thrust-belt deformation along the eastern edge of the Disturbed Belt in the Helena salient of Montana has not been well dated owing to a lack of syn- or postorogenic strata. *In situ* paleomagnetic data from alkaline intrusions exposed in the easternmost folds of the salient in the northern Crazy Mountains Basin, previously described as pre-, syn-, or posttectonic with respect to deformation, are well grouped (Dec. = 343°, Inc. = 61°, α_{95} = 5.0°, k = 46, n = 16 sites); a negative fold test is significant at minimally 95% confidence. Intrusion and magnetization acquisition thus postdate fold and thrust deformation. K-Ar age determinations of selected intrusions range from 52 to 48 Ma. Paleomagnetic and isotopic age data, combined with stratigraphic information, indicate that folding in the northern Crazy Mountains Basin is middle to late Paleocene in age. This age is in agreement with suggested dates for deformation in the northern Montana Disturbed Belt but is recognizably older than the youngest episodes of frontal deformation in the Utah-Idaho-Wyoming salient. Data from this study and existing structural and stratigraphic information demonstrate that deformation in the overthrust belt and foreland provinces of southwest Montana overlapped temporally and spatially, ranged from Late Cretaceous to earliest Eocene in age, and progressed from west to east through time.

INTRODUCTION

The geology of southwest Montana is characterized by two distinct styles of Laramide-age (Late Cretaceous to early Tertiary) deformation: (1) thin-skinned décollement-style folds and thrust faults of the overthrust belt and (2) basement-involved folds and faults of the Rocky Mountain foreland province (Fig. 1). In general, deformation in both provinces is considered to have ranged from Late Cretaceous to middle or late Eocene time (McMannis, 1965; Roberts, 1972; Schmidt and Hendrix, 1981) and is thought to have progressed from west to east through time (Schmidt and Garihan, 1983). Unfortunately, in many areas, precise dating of thrust movements is precluded by an absence of preserved overlying syn- or postorogenic strata. Where synorogenic strata are present, their source areas and/or ages are in many cases equivocal. In contrast, the timing of individual thrust movements in much of the Utah-Idaho-Wyoming salient and adjacent foreland uplifts of the central Rocky Mountains has been highly refined (Dorr and others, 1977; Wiltschko and Dorr, 1983; Gries, 1983). Alternative dating methods are necessary in many areas in order to determine the timing and progression of defor-

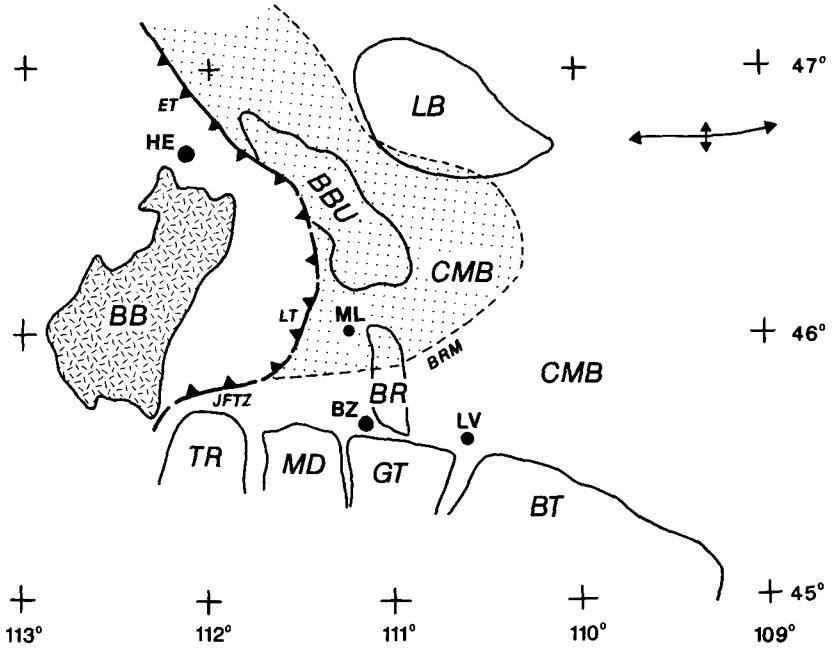


Figure 1. Map showing the distribution of thin-skinned deformation of the overthrust belt and major foreland uplifts in southwest Montana. Stippled pattern represents the area of the Montana Disturbed Belt. Symbols: BB = Boulder batholith; TR = Tobacco Root Mountains; MD = Madison Range; GT = Gallatin Range; BT = Beartooth Mountains; CMB = Crazy Mountains Basin; BR = Bridger Range; BBU = Big Belt uplift; LB = Little Belt Mountains; JFTZ = Jefferson Canyon transverse zone; LT = Lombard thrust; ET = Eldorado thrust; BRM = Battle Ridge monocline; and BZ = Bozeman, LV = Livingston, ML = Maudlow, and HE = Helena. Figure modified from Woodward (1981).

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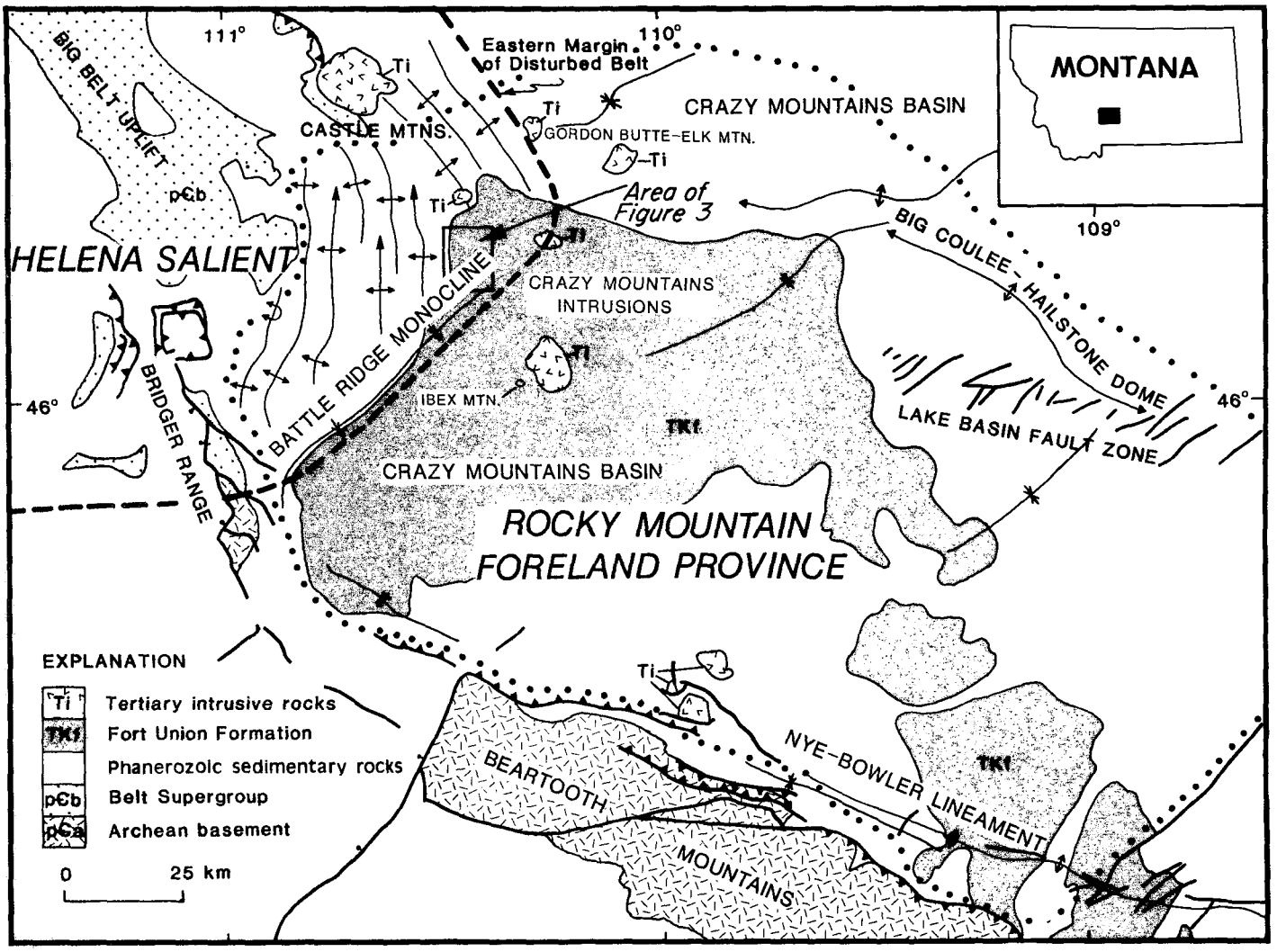


Figure 2. Tectonic map of the Crazy Mountains Basin and surrounding uplifts. Map shows the intersection of the Helena salient of the Disturbed Belt and the Rocky Mountains foreland province. Dotted line indicates the approximate margins of the Crazy Mountains Basin. Modified from Roberts (1972).

mation in the Helena salient and adjacent foreland of southwest Montana.

A series of Tertiary intrusions, variously described as pre-, syn-, or posttectonic with respect to local deformation (for example, Larsen and Simms, 1972; Roberts, 1972; Tanner, 1949), are exposed in the easternmost folds of the Helena salient and foreland anticlines of the Crazy Mountains Basin (Fig. 2). Although the intrusions cut lowermost Tertiary strata, their absolute age is poorly known. Harlan and Lageson (1983) suggested that paleomagnetic fold tests on arcuate sills in the Robinson anticline area of the northern Crazy Mountains Basin, in combination with isotopic (K-Ar) dating, may bracket the age of fold and thrust deformation. The paleomagnetic and isotopic data presented herein more accurately constrain the timing of deformation along the eastern margin of the Helena salient. These data are then integrated with observations bearing on the timing of deformation from other areas of southwest Montana in order to document the age and progression of Laramide-age tectonism in the Helena salient and adjacent foreland uplifts of southwest Montana.

GEOLOGIC SETTING

The Crazy Mountains Basin is a complex Late Cretaceous to Eocene broken foreland basin in west-central Montana which overlaps the intersection of thin-skinned, décollement-style fold and thrust structures of the Montana Disturbed Belt (Woodward, 1981; Garrett, 1972) and basement-involved folds and thrust faults of the Rocky Mountain foreland province (Schmidt and Garihan, 1983; Gries, 1983) (Fig. 2). The Battle Ridge monocline along the southeastern margin of the Helena salient in the Crazy Mountains Basin locally defines this intersection and is a northeast extension of the right-lateral Jefferson Canyon transverse zone (Fig. 1) (Schmidt and Garihan, 1983; Roberts, 1972).

Disturbed Belt structures along the western margin of the Crazy Mountains Basin consist of north-south–trending, strongly asymmetrical, east-verging folds and west-dipping thrust faults. To the east, near the center of the basin, Disturbed Belt folds are characterized by slightly asymmetrical, open structures which are convex eastward in map view (Fig. 2) and are probably cored by blind thrusts in the subsurface (Garrett, 1972; Woodward, 1981). Along the southern margin of the salient, many folds are deflected westward at their southern end near the trace of Battle Ridge monocline. Battle Ridge monocline

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is thought to be the surface expression of an oblique-slip lateral ramp which soles into a regional décollement at or near the base of the Precambrian Belt Supergroup (Woodward, 1981; Garrett, 1972). The study area focuses on the southern end of Robinson anticline, an open, east-verging fold along the eastern margin of the salient (Fig. 2). Robinson anticline trends northwest in the Castle Mountains and approximately north-south at its southern termination, where structural analysis of poles to bedding and sill contacts defines a 25°, S. 2° E. plunge. As in the case of folds to the west, it is probably cored by a blind thrust at depth. Numerous concordant intrusions are exposed on the south-plunging nose of Robinson anticline (Simms, 1966) (Fig. 3); to the north of the study area, the fold is intruded by the Comb Creek stock and dike swarm (Sims, 1967; Fink, 1975).

Foreland province structures in the Crazy Mountains Basin include folds with diverse trends in the basin proper and northwest-trending, basement-involved and out-of-the-basin thrust faults along the northern margin of the Beartooth and Gallatin uplifts (Fig. 2). In addition, foreland deformation includes left-lateral transform movement along the northwest-trending Nye-Bowler lineament (Wilson, 1936; Geissman and Mogk, 1986) and Lake Basin fault zone (Chamberlin, 1919).

The stratigraphy of the Crazy Mountains Basin consists of Paleozoic and Mesozoic carbonate, sandstone, and shale exposed along the basin margins and surrounding uplifts and of Upper Cretaceous–middle Paleocene (?) clastic and volcanic basin fill (Fig. 2). In addition, Paleozoic and Mesozoic sedimentary rocks in the northern part of the basin overlie a thick sequence of Proterozoic carbonate, sandstone, and argillite of the Belt Supergroup. Upper Cretaceous to lower Tertiary synorogenic sandstones and shales of the Fort Union Formation are deformed at Robinson anticline.

Subalkaline and alkaline stocks, sills, and dikes are present near the approximate center of the basin (Fig. 2). Two distinct, spatially and temporally overlapping magma series are present (Dudas and Eggler, 1984; Dudas and others, 1987): (1) sodium-rich, strongly alkaline mafic syenites, syenites, and trachytes of the Comb Creek stock and dike swarm, Gordon Butte–Elk Mountain–Coffin Butte sills, and the Robinson anticline intrusive complex (RAIC) (Simms, 1966; Larsen and Simms, 1972; Fink, 1975) and (2) subalkaline rocks consisting of potassium-rich basalts, diorites, and granodiorites of the Big Timber and Loco Mountain stocks and associated dike swarms (Larsen and Simms, 1972). Concordant intrusions of the RAIC area (Fig. 3) and in adjacent foreland folds to the east were considered to have been emplaced either pre- or syntectonically (Simms, 1965, 1966; Larsen and Simms, 1972; Wolff, 1938; Sims, 1967; Tanner, 1949; Bowen, 1918) based on (1) over-all map patterns suggestive of folded strata; (2) the tendency for some sills to thicken along fold hinges rather than along limbs, possibly suggesting synfolding intrusion (Simms, 1966); and (3) a general restriction of large sills to troughs or crests of folds (Bowen, 1918). The Comb Creek, Loco Mountain, and Big Timber stocks and dikes clearly cut fold trends and hence postdate deformation (Sims, 1967; Fink, 1975; Roberts, 1972). Previous attempts to determine the age of folding relative to intrusion by structural and paleomagnetic methods were inconclusive (Simms, 1966). Although alkaline igneous activity is not characteristically associated with compressional deformation, syntectonic alkaline intrusions of compositions similar to those of the northern Crazy Mountains Basin occur in the Moine thrust zone, Scotland (Coward, 1986). In addition, lower Tertiary (Paleocene?) potassium-rich alkaline volcanic rocks of the Adel Mountains field in the Disturbed Belt northwest of Helena, Montana, have been folded and cut by thin-skinned thrusts of Laramide age (Schmidt, 1978). That other alkaline rocks within the northern Disturbed Belt are demonstrably pretectonic suggests that intrusions in the

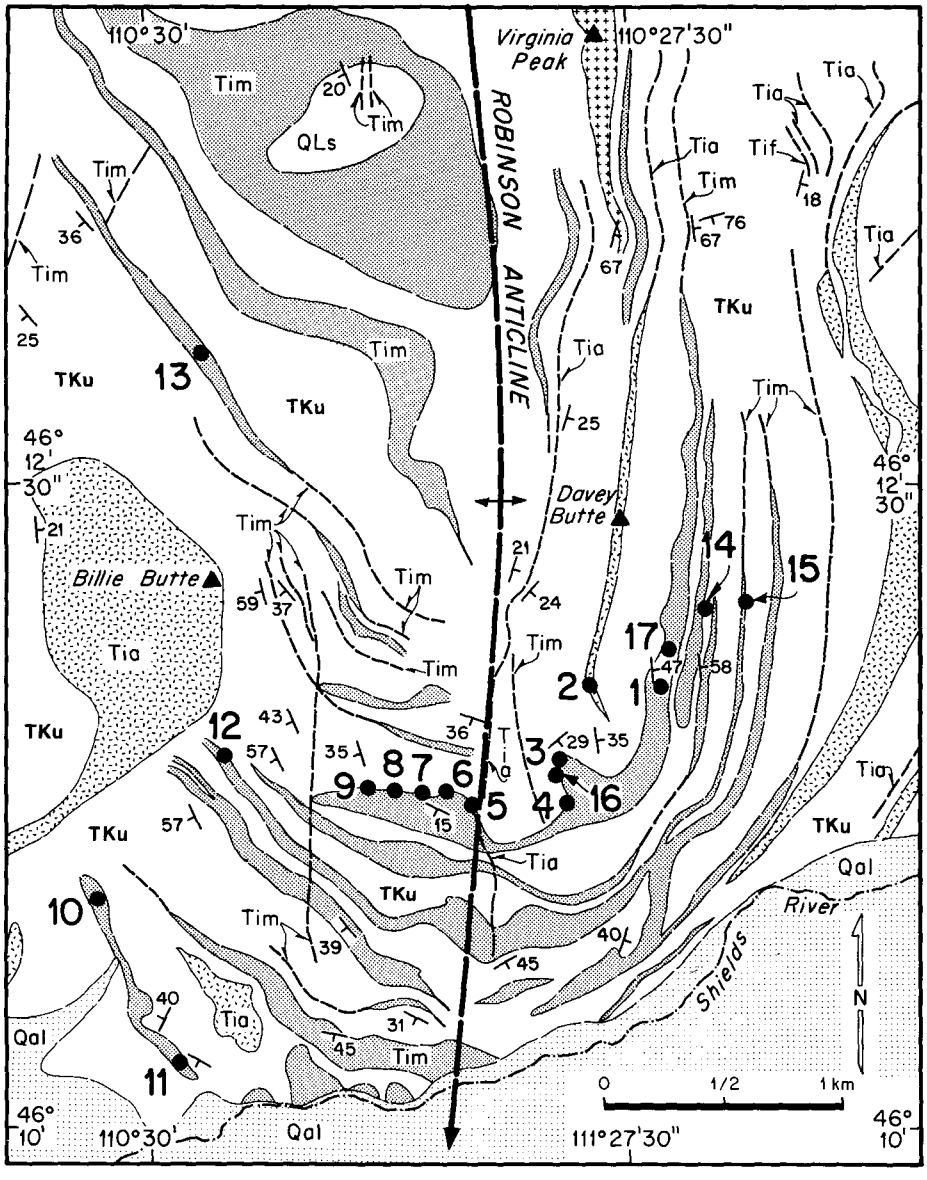


Figure 3. Geologic map of the southern part of Robinson anticline, northern Crazy Mountains Basin. Symbols: QLs = landslide deposits, Qal = alluvium and stream terrace deposits, Tia = trachyte dikes and sills, Tif = feldspathoidal syenite dikes and sills, Tim = mafic nepheline syenite dikes and sills, TKu = Fort Union Formation sandstones and shales, and numbered circles = paleomagnetic sampling sites. Modified from Simms (1966) and Larsen and Simms (1972).

Crazy Mountains basin may have been pre- or syntectonic as well.

PALEOMAGNETISM

Methods

Independently oriented samples (117 samples) were collected from 17 sites in mafic nepheline syenite (16 sites) and trachyte sills (1 site) distributed around the south-plunging nose of Robinson anticline (Fig. 3). Evaluation of the age of intrusion relative to the folding event was accomplished using the principles outlined in Graham's (1949) fold test; a recent example applied to intrusions and their relationship to folding in the Colorado Plateau–Basin and Range transition zone is described by Sandberg and Butler (1985).

Application of a fold test to concordant igneous rocks requires accurate determination of the structural attitude of the intrusions. Because sill/bedding contacts are generally poorly exposed and the sills do not in most cases contain internal structures which allow determination of their attitude, sampling was restricted to sites where the attitude of adjacent host strata and/or sill contacts could readily be measured and tilt corrections applied. Samples were collected as oriented blocks (3–7 samples/site, 12 sites) or using a portable drill (7–8 samples/site, 5 sites). Orientations were obtained using a magnetic compass and were checked by sun compass or backsighting techniques.

Measurements of the natural remanent magnetization (NRM) were made using a Schonstedt SSM-1a spinner magnetometer. Progressive alternating field (AF), thermal, and AF followed by thermal demagnetization techniques were employed to resolve all remanent magnetization components. Components were identified using orthogonal vector diagrams (Zijderveld, 1967). Vector subtraction (stable endpoint) and/or remagnetization circle techniques (Bailey and Halls, 1984) were used to calculate directions of remanent magnetization characteristic of each site. Saturation magnetization versus temperature (Curie temperature) measurements, isothermal remanent magnetization (IRM) acquisition tests, and petrographic examination were used to identify significant remanent magnetization carriers. Anisotropy of magnetic susceptibility (AMS) measurements were made using a Sapphire SI-2 induction coil system.

Magnetic Properties

IRM curves of the mafic nepheline syenite (Fig. 4) typically show saturation in fields less than 0.150 T and low coercivity of remanence (Hcr) values (0.075–0.045 T) in backfield demagnetization. Results from contact hornfels rocks show saturation at higher fields (0.30–0.40 T) and Hcr of 0.085 T. Curie temperatures ranged from 550–570 °C, and curves were within 5% of reversible (Fig. 5). Together with demagnetization results, these data suggest that fine-grained, homogeneous, low-Ti titanomagnetite is the principal remanence carrier in the mafic nepheline syenites of the RAIC and that both titanomagnetite and hematite are important magnetic phases in the baked contact zones.

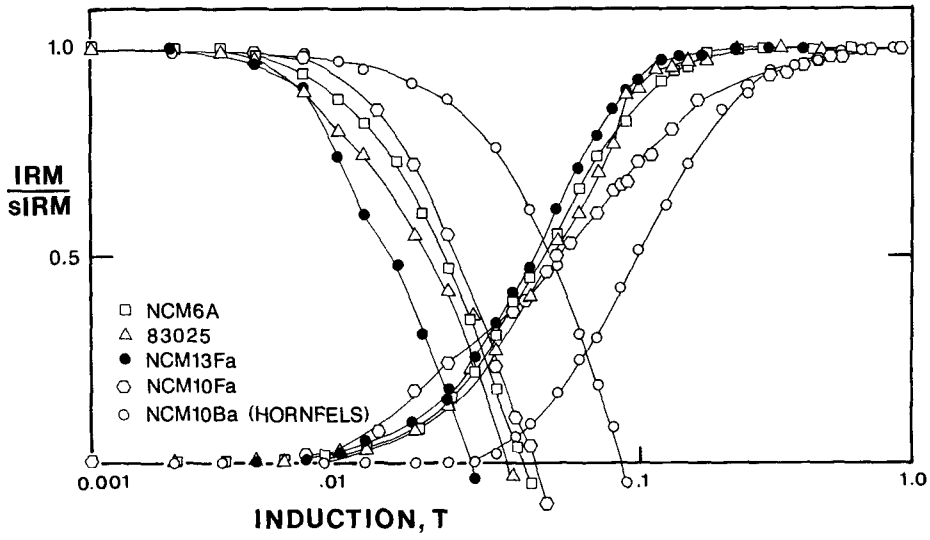


Figure 4. Isothermal remanent magnetization (IRM) acquisition (positive field) and backfield demagnetization (negative field) curves from mafic nepheline syenite and hornfels samples from the RAIC. Curves are normalized to saturation IRM (sIRM).

Particularly near contacts, sills of the RAIC display a preferred orientation of silicate (mainly pyroxene) minerals, as noted by Simms (1966). To evaluate potential effects of pronounced silicate mineral anisotropy on the acquisition of a thermoremanent magnetization (TRM) in the RAIC sills, anisotropy of magnetic susceptibility (AMS) data were collected from samples from six sites dispersed around Robinson anticline. Using modified Flinn-type diagrams (for example, Hrouda, 1982) where k_1/k_3 is plotted versus k_2/k_3 (k_1, k_2, and k_3 = maximum, intermediate, and minimum axes, respectively, of a triaxial susceptibility ellipsoid), we observed sub-equal populations of oblate and prolate AMS ellipsoids, suggesting the lack of a well-defined magnetic lineation or foliation (Fig. 6). Observed percentages of anisotropy [defined as $(k_1 - k_3)/k_{mean}$] are in most cases less than 3.5%. Consequently, we believe that the possibility of a silicate mineral fabric strongly influencing TRM acquisition in the RAIC sills is remote. We note, however, that anisotropy of TRM and

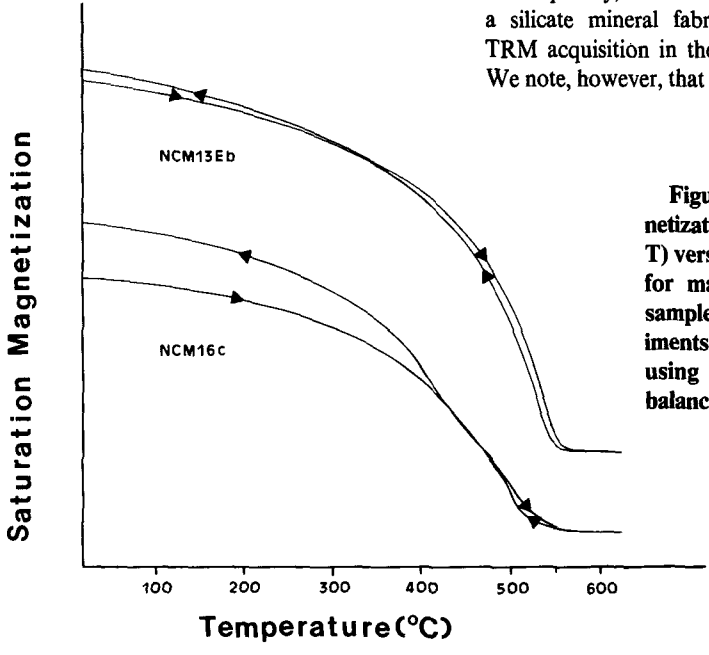


Figure 5. Saturation magnetization (induction of ~0.25 T) versus temperature curves for mafic nepheline syenite samples from RAIC. Experiments were conducted in air using a horizontal Curie balance.

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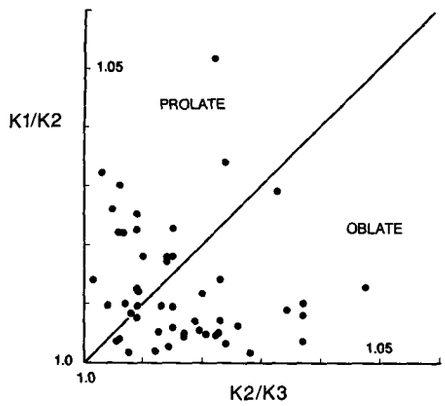


Figure 6. Plot of anisotropy of magnetic susceptibility data (k_1/k_2 versus k_2/k_3) for RAIC samples. Line of unit slope separates the prolate and oblate ellipsoid fields.

magnetic susceptibility are not necessarily similar, differences being in part strongly controlled by the domain state of the magnetic phases (for example, Stephenson and others, 1986). At each site, the mean directions of the minimum principal susceptibility axes (k_3) (perpendicular to magnetic foliation) are well defined and generally perpendicular to sill/bedding contacts.

Results

NRM intensities of mafic syenites from Robinson anticline range from 0.1 to 3.0 A/m. Demagnetization by AF and thermal methods was straightforward. In most cases, a single magnetization component was isolated, recognized by at least three colinear points which showed a monotonic decay to the origin of a vector diagram (Fig. 7). Superimposed secondary magnetizations, in most cases characterized by due-northerly directions and moderate to steep positive inclinations and hence of recent origin, were removed by peak inductions of 10–20 mT and temperatures of 180–300 °C. No other secondary(?) magnetization of consistent direction was observed.

Samples which exhibited a more complicated behavior contain two or more magnetization components of overlapping coercivity and/or laboratory unblocking temperature spectra and have NRM intensities greater than 10 A/m. They probably contain an IRM caused by lightning strikes. Specimens for which a well-defined component of magnetization could not be isolated, or those which differed from the calculated site mean by greater than two standard deviations, were rejected from statistical analysis. The entire sample collection of trachyte from site 2 was rejected owing to unstable behavior during demagnetization.

In situ site-mean directions are shown in Figure 8 and paleomagnetic data in Table 1. Two antipodal groups of site-mean directions were obtained (Table 1); reversed-polarity data have been inverted to common north-seeking, positive inclination directions in Figure 8. Reversed-polarity data include samples of the baked contact zone adjacent to site 10. Interestingly, sills at lower stratigraphic levels on the RAIC are consistently of reversed polarity, whereas normal-polarity sites are found in sills from higher stratigraphic levels. The isotopic age data (discussed below) lead us to suggest that sill intrusion spanned polarity chrons 22 and 21 (ca. 51–48 Ma) (Harland and others, 1982).

Given uncertainties regarding the kinematics of fold plunge development (for example, MacDonald, 1980), structural corrections to assess the age of remanent magnetization acquisition were applied in two ways. In the first method, site directions were restored to the paleohorizontal by simple correction for bedding tilt about the line of strike of the host strata. Alternatively, fold plunge was restored to horizontal followed by bedding tilt corrections as described above (Phillips, 1971; Ramsay, 1961). Comparison of *in situ* and corrected site-mean directions (Fig. 7 and Table 1) shows that dispersion of site-mean directions increases significantly, regardless of the structural correction method employed. Incremental step-wise restoration of the fold to the assumed prefolding configuration (for example, Kent and Opdyke, 1986) results in a progressive increase in dispersion of site-mean directions and a systematic decrease in Fisher's precision parameter k. Although we are aware of the inherent problems with incremental fold tests (for example, Kodama, 1986; van der Pluijm, 1987) and that intrusion of the sills may have locally modified the attitude of host strata, we believe that our data indicate that sills of the RAIC were injected subsequent to compressional folding. Statistical evaluation using McElhinny's (1964) f-ratio test and McFadden and Jones' (1981) more rigorous method demonstrates that the fold test is negative and significant, minimally at the 95% confidence level. TRM acquisition, and hence sill intrusion, thus occurred subsequent to Laramide folding.

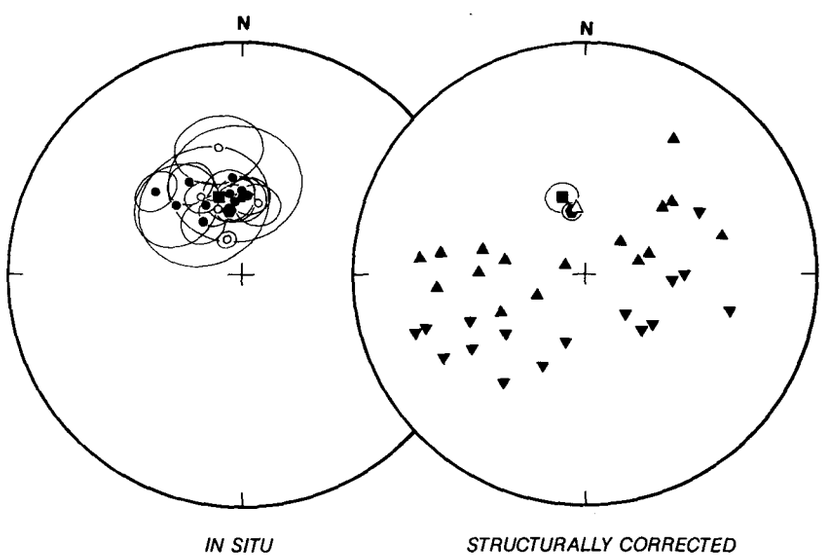


Figure 7. Representative orthogonal demagnetization diagrams (Zijderveld, 1967) for Robinson anticline intrusive-complex samples. Peak alternating fields (in milliTesla) or temperatures (in degrees Celsius) are given along the vertical projection (open symbols) of the remanent magnetization; horizontal projections are denoted by closed symbols.

ISOTOPIC DATING

Samples for K-Ar dating were collected from the Comb Creek stock and dike swarm and from sills of the RAIC. Both intrusive centers are exposed along the hinge of Robinson anticline and, as stated previously, the RAIC has been considered to be pre- or synfolding and the Comb Creek stock, to the north of the study area, to be postfolding. Data from hornblende and biotite separates are listed in Table 2. In addition, a date from a mafic syenite sill at Ibex Mountain (Theralite Butte) (Fig. 2) is reported. This intrusion is similar in composition to those of the RAIC but is south of the Disturbed Belt. Dates obtained from the alkaline intrusions range from 52–48 Ma and are statistically indistinguishable at the 95% confidence level. These dates are in good agreement with a recently reported Rb-Sr mineral isochron age of 48 ± 2 Ma for Crazy Mountains alkaline intrusions reported by Dudas and others (1987). This early middle Eocene age for intrusive activity is comparable to that of alkaline igneous activity else-

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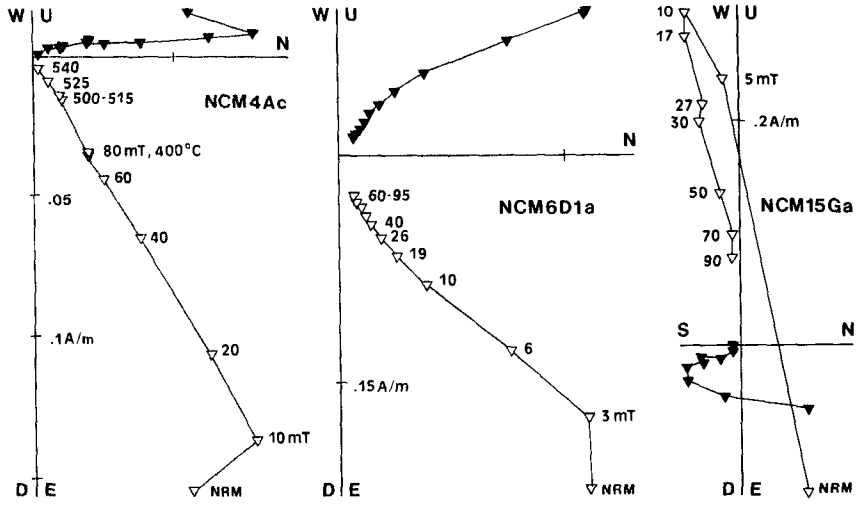


Figure 8. Equal-area projection of *in situ* and structurally corrected site mean directions and their associated α_{95} cones of confidence. Upright triangles are site means restored to the paleohorizontal by simple bedding tilt correction; inverted triangles are site means after correction for fold plunge and bedding tilt. Open circles denote site mean directions which have been inverted to north-seeking declinations. For clarity, *in situ* group mean and reference directions are portrayed in the structurally corrected diagram. Square is observed mean direction from RAIC sites; octagon is observed mean direction of Diehl and others (1983) for lower Tertiary intrusions in central Montana; open triangle is expected 50 Ma direction for central Montana (Dec. = 349°, Inc. = 67°, α_{95} = 4°; Irving and Irving, 1982).

where in central Montana (Marvin and others, 1973, 1980; Marvin and Dobson, 1979).

DISCUSSION

Robinson Anticline

Paleomagnetic data from sills of the RAIC demonstrate that their emplacement postdated deformation. Our field observations suggest that sill intrusion was localized at mudstone-shale/sandstone contacts, although this relationship is obscured in most places by poor outcrop exposures of the contact zones. The outcrop distribution appearance, although suggestive of pre- or synfolding intrusion, is in fact due to postfolding emplacement. Where structural control was not favorable for sill intrusion, that is, in areas of nearly flat-lying sedimentary rocks, dike emplacement occurred (Simms, 1966).

Paleomagnetic data also indicate NRM acquisition over a time interval of dual polarities, more likely chron 22 or, possibly, chron 21, although specific chron assignment is not possible owing to analytical error inherent in the isotopic ages. The presence of dual polarities in RAIC units is significant because previous field data from the northern Crazy Mountains (Fink, 1975) were exclusively of normal polarity. Our *in situ* group mean direction (Dec. = 343°, Inc. = 62°) is statistically indistinguishable, using the method of McFadden and Lowes (1981), at a 95% confidence level from that reported by Diehl and others (1983) for lower Tertiary (Paleocene-Eocene) intrusions in central Montana (Dec. = 350°, Inc. = 65°, extrapolated from the Diehl and others, 1983, 44–54 Ma pole position to Crazy Mountains Basin coordinates), which lie outside the area of obvious Laramide deformation.

Because RAIC intrusions postdate deformation, they cannot be used to place older age limits on folding, as originally proposed by Harlan and Lageson (1983). Isotopic data from these intrusions, as well as those of the Comb Creek stock and dike swarm, place a younger age limit of early middle Eocene (ca. 50 Ma) on fold and thrust deformation in the northern Crazy Mountains Basin. Mafic intrusions of similar composition which are exposed in foreland folds immediately east of the Disturbed Belt (for example, Coffin Butte, Little Elk dome, and so on) (Fig. 2) are probably posttectonic as well.

An older age limit for deformation is provided by the youngest sedimentary unit involved in Disturbed Belt folding at Robinson anticline. This unit is the synorogenic Fort Union Formation, which has been dated by paleontologic and palynologic methods as Late Cretaceous to middle Paleocene in age (Roberts, 1972). Determination of the age and source area of the Fort Union Formation in the study area, however, is

TABLE 1. SUMMARY OF SITE MEAN PALEOMAGNETIC DATA FROM THE ROBINSON ANTICLINE INTRUSIVE COMPLEX, NORTHERN CRAZY MOUNTAINS, MONTANA

Site	Polarity	*in situ*		Corrected*		Corrected†		Attitude(°)		α_{95}	k	R	N/No
		D(°)	I(°)	D(°)	I(°)	D(°)	I(°)						
1	Normal	351	61	052	52	050	54	010	35	8.7	54	5.93	6/7
2	"	components not separated											0/5
3	Normal	354	55	047	74	050	73	055	29	23.0	13	2.84	3/4
4	Normal	001	63	073	66	073	66	042	31	9.0	61	3.95	4/5
5	Normal	332	62	258	69	294	82	076	25	22.0	10	3.90	4/4
6	Normal	314	47	278	38	278	38	130	38	7.2	75	4.95	5/5
7	Normal	316	56	271	52	270	52	120	32	15.0	22	3.86	4/5
8	Normal	330	52	275	49	282	52	125	34	8.4	47	5.94	6/6
9	Normal	324	67	272	52	245	57	115	38	8.2	12	2.99	3/3
10	Reversed	160	−65	268	42	265	36	145	50	6.0	63	7.89	8/8
11	Reversed	194	−64	252	69	244	71	124	40	8.0	40	6.85	7/8
12	Reversed	152	−59	276	28	275	39	150	57	5.0	96	7.93	8/8
13	Normal	354	64	280	61	279	62	140	35	7.0	51	7.86	8/8
14	Reversed	170	−43	034	31	34	30	000	53	14.0	20	4.80	5/6
15	Reversed	157	−76	073	35	75	38	000	58	3.0	280	7.98	8/8
16	Normal	005	62	077	70	77	71	055	29	7.0	67	5.93	6/7
17	Normal	000	60	049	49	52	50	010	35	4.0	155	7.96	8/8
Note: means; 16 sites,		343° 61° R = 15.675 k = 46.14 α_{95} = 4.0°		319° 78° R = 12.871 k = 7.794 α_{95} = 16°		197° 67° R = 12.50 k = 4.287 α_{95} = 17°							

D = declination, I = inclination; α_{95} = semi-angle of 95% confidence level; N/No = ratio of samples used in calculation to samples collected and measured; k = precision parameter (Fisher, 1953); and R = resultant vector.
*Simple bedding tilt correction.
†Plunge and bedding tilt corrections.

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TABLE 2. ISOTOPIC (K-Ar) DATES FROM SELECTED CRAZY MOUNTAINS INTRUSIONS

Sample no.	Field no.	Igneous center	Locality	Material dated/rock type	K_2O (%)	$^{40}Ar^*$ (ppm)	Age (Ma)
1	84CC12	Comb Creek stock	46°18'5"N, 110°30'W	Biotite/neph. syenite	6.923	0.02336	48.0 ± 1.9
2	83VPCC1	Comb Creek dike	46°17'52"N, 110°26'49"W	Biotite/mafic neph. syenite	6.453	0.02443	52.6 ± 2.0
3	83VP65	RAIC sill	46°11'14"N, 110°29'12"W	Biotite/mafic neph. syenite	7.380	0.02589	50.6 ± 1.9
4	83VP75	RAIC sill	46°11'19"N, 110°27'49"W	Biotite/mafic neph. syenite	6.718	0.02328	49.3 ± 4.6
5	82VP2	RIAC sill	46°12'10"N, 110°27'30"W	Hornblende/trachyte	0.167	0.00584	48.8 ± 4.6
6	008	Ibex Mtn. sill(?)	45°59'12"N, 110°27'1"W	Biotite/mafic neph. syenite	6.5116	0.02309	50.5 ± 0.6

Note: $\lambda_\beta = 4.962 \times 10^{-10}$/yr, $\lambda_e = 0.581 \times 10^{-10}$/yr; samples 1–5 dated by Krueger Enterprises, Cambridge, Massachussets; sample 6 collected by R. A. Chadwick (deceased) and dated by L. W. Snee, Isotope Laboratory, Ohio State University, Columbus, Ohio.

problematic. Sims (1964), on the basis of lithologic and paleontologic similarities, tentatively correlated rocks in the study area with strata east of the Crazy Mountains, described previously as the Melville Formation (Simpson, 1937). The Melville Formation is now identified as the Tongue River member of the Fort Union Formation of middle to late Paleocene age (Roberts, 1972). If this correlation is correct and if the units are of similar age, then fold and thrust deformation is post-middle or late Paleocene and pre–early middle Eocene in age. Unfortunately, stratigraphic relations between Robinson anticline strata and dated Fort Union sections to the south and west have not been established. If sedimentary units in the study area belong to the lower strata of the Fort Union Formation, then only an older age limit of Late Cretaceous or early Paleocene age for deformation may be inferred. Additional stratigraphic work is needed to more accurately resolve the age of thrust deformation in this area.

Paleomagnetic and isotopic age data, combined with pre-existing stratigraphic studies, thus indicate that thin-skinned fold and thrust, as well as nearby foreland deformation in the northern Crazy Mountains Basin, is at most early Paleocene in age and may be as young as middle or late Paleocene, to possibly earliest Eocene. Although we cannot rule out an earliest Eocene age as a younger limit for deformation, we believe that deformation had ceased prior to early Eocene time. A middle or late Paleocene age for folding is in reasonable agreement with the timing of deformation of similar structural style along the eastern margin of the northern Disturbed Belt (Mudge, 1982) but is recognizably older than frontal deformation along the eastern margin of the Utah-Idaho-Wyoming salient.

Timing of Laramide Tectonism in Southwest Montana

Although precise dating of motions of individual thrusts in southwest Montana is made complicated by the absence or equivocal nature of synorogenic strata in this region, data and observations bearing on the timing of deformation of Robinson anticline and other areas can be used to generally assess the age and progression of Laramide tectonism in southwest Montana (area shown in Fig. 1). As stated previously, two distinct styles of deformation (foreland versus fold and thrust belt structures) overlapped both temporally and spatially during Late Cretaceous–early Tertiary time. The two deformation styles, however, represent differing mechanical responses to horizontal compression, which began in Late Jurassic or Early Cretaceous time, in relation to plate-tectonic processes along the western margin of North America. The timing of deformation in each of the two provinces will be considered separately.

In the foreland province (Fig. 1), uplift of the Precambrian-cored Tobacco Root Mountains in Late Cretaceous time has been documented by Schmidt and Garihan (1983). They place the major period of movement on northwest-trending faults as occurring between 70 and 75 Ma, based on crosscutting relationships and isotopic dates derived from the Tobacco Root and Boulder batholiths (preorogenic) and the Hell Canyon pluton (postorogenic). To the south and east of the Tobacco Root Mountains, in the Madison Range, Tysdal and others (1986) have dated motion on basement-involved northwest-trending faults as 79–68 Ma on the basis of crosscutting relationships and isotopic dating. Here, the Upper Cretaceous (Campanian) Livingston Group, dated by K-Ar (welded ash-flow tuffs) and palynological methods as ca. 79 Ma, is involved in the Laramide deformation; dacite dikes and sills yielding ^{40}Ar-^{39}Ar dates of 68 Ma clearly postdate these structures.

Laramide uplift of the Beartooth Range is constrained by synorogenic strata and postorogenic igneous activity as latest Cretaceous to possibly earliest Eocene in age. Synorogenic sedimentation is recorded by the Fort Union Formation along the northern and eastern margins of the Beartooth Mountains in the Crazy Mountains Basin and the northern Bighorn Basin. Dutcher and others (1986), on the basis of sedimentological studies of Paleocene strata in the Bighorn Basin, have suggested that uplift of the Beartooth Range was a two-stage process, with major uplift culminating during the late Paleocene. Laramide deformation had clearly ceased prior to eruption of the areally extensive lavas of the Absaroka-Gallatin field beginning at approximately 52 Ma (Chadwick, 1969, 1972; Smedes and Prostka, 1972; Roberts, 1972).

In the fold and thrust belt, thin-skinned deformation along the western margin of the Helena salient (Fig. 1) occurred in Late Cretaceous time as evidenced by thrusts which cut both the Boulder batholith and Elkhorn Mountains volcanics (ca. 75–85 Ma) and by the truncation of foreland folds of the Tobacco Root uplift by thrusts of the right-lateral Jefferson Canyon transverse zone (Schmidt and Garihan, 1983). Northwest of Helena, the Eldorado-Lombard thrust has been cut by a dike which yields a K-Ar age of 58.3 Ma; this places an upper age limit of latest Cretaceous to early Paleocene for movement of the Elkhorn thrust plate (Mehnert and Schmidt, 1971). In the Maudlow area, in the central part of the salient, Harlan (1986) has suggested that thin-skinned thrusts structurally elevated Mesozoic, Paleozoic, and Precambrian strata in latest Cretaceous time, thereby providing a source area for the coarse conglomerates of the Fort Union Formation exposed along the eastern flank of the Bridger Range (Roberts, 1972). These conglomerates were subsequently deformed by thrust faults now exposed in the northern Bridger Range and along the Battle Ridge monocline. Along the eastern margin of the Helena salient, our data indicate that deformation was at most early Paleocene and possibly as young as middle to late Paleocene in age.

SUMMARY AND CONCLUSIONS

Paleomagnetic data from this study demonstrate that lower Tertiary sills of the RAIC

which intrude frontal folds of the Disturbed Belt in the Crazy Mountains Basin are post-tectonic with respect to Laramide deformation. This result, combined with new isotopic dates and preexisting stratigraphic studies, is used to infer a middle to late Paleocene age for Disturbed Belt deformation along the leading edge of the Helena salient. Data from this study, as well as existing structural and stratigraphic information, indicate that Laramide tectonism in the Helena salient and adjacent foreland uplifts of southwest Montana spanned the period from Late Cretaceous to possibly earliest Eocene time. Deformation in both the overthrust belt and foreland provinces progressed roughly from west to east through time, although at least locally, motion on individual thrusts may have overlapped. To the west, along the western margin of the Helena salient, foreland deformation preceded thin-skinned or overthrust deformation, in contrast to the progression of tectonism in the central Rocky Mountains (Wiltschko and Dorr, 1983; Dorr and others, 1977; Gries, 1983). In the central Crazy Mountains Basin, the timing of Disturbed Belt folding (this study) and foreland uplift of the Beartooth Range may have been largely synchronous, within the resolution of available data. Deformation in both the foreland and the Disturbed Belt had clearly ended by the early middle Eocene; this age is significantly younger than the age of cessation of Laramide tectonism in the central and southern Rocky Mountains.

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REFERENCES CITED

Bailey, R. C., and Halls, H. C., 1984, Estimate of confidence in paleomagnetic directions derived from mixed remagnetization circle and direct observational data: Journal of Geophysics, v. 54, p. 174–182.

Bowen, C. F., 1918, Anticlines in a part of the Musselshell Valley, Musselshell, Meagher, and Sweetgrass Counties, Montana: U.S. Geological Survey Bulletin 691, p. 185–209.

Chadwick, R. A., 1969, The northern Gallatin Range, Montana: Northwestern part of the Absaroka-Gallatin volcanic field: University of Wyoming Contributions to Geology, v. 8, p. 150–166.

——1972, Volcanism in Montana: Northwest Geology, v. 1, p. 1–20.

Chamberlin, R. T., 1919, A peculiar belt of oblique faulting: Journal of Geology, v. 27, p. 602–613.

Coward, M. P., 1986, The thrust structures of southern Assynt, Moine thrust zone: Geological Magazine, v. 122, p. 595–607.

Diehl, J. F., Beck, M. E., Jr., Beske-Diehl, S., Jacobsen, D., and Hearn, B. C., Jr., 1983, Paleomagnetism of the Late Cretaceous–early Tertiary north-central Montana alkalic province: Journal of Geophysical Research, v. 88, p. 10593–10609.

Dorr, J. A., Spearing, D. R., and Steidtmann, J. R., 1977, Deformation and deposition between a foreland uplift and an impinging thrust belt: Hoback Basin, Wyoming: Geological Society of America Special Paper 177, 82 p.

Dudas, F. O., and Eggler, D. H., 1984, Geochemistry of igneous rocks in the Crazy Mountains, Montana: Geological Society of America Abstracts with Programs, v. 16, p. 494–495.

Dudas, F. O., Carlson, R. W., and Eggler, D. H., 1987, Regional Middle Proterozoic enrichment of the subcontinental mantle source of igneous rocks from central Montana: Geology, v. 15, p. 22–25.

Dutcher, L.A.F., Jobling, J. L., and Dutcher, R. R., 1986, Stratigraphy, sedimentology and structural geology of Laramide synorogenic sediments marginal to the Beartooth Mountains, Montana and Wyoming: Montana Geological Society–Yellowstone Bighorn Research Association 1986 Field Conference Guidebook, p. 33–52.

Fink, R. A., 1975, The geology and igneous petrology of the Comb Creek dike swarm, Crazy Mountains, Montana [M.S. thesis]: Cincinnati, Ohio, University of Cincinnati, 174 p.

Fisher, R. A., 1953, Dispersion on a sphere: Royal Society of London Proceedings, v. A217, p. 295–305.

Garrett, H. L., 1972, Structural geology of the Crazy Mountains Basin: Montana Geological Society, Annual Field Conference, 21st, Guidebook, p. 185–195.

Geissman, J. W., and Mogk, D. W., 1986, Late Archean tectonic emplacement of the Stillwater Complex along reactivated basement structures, northern Beartooth Mountains, southern Montana, U.S.A.: International Conference on Basement Tectonics, 6th, Proceedings, p. 25–44.

Graham, J. W., 1949, The stability and significance of magnetism in sedimentary rocks: Journal of Geophysical Research, v. 54, p. 131–167.

Gries, R., 1983, North-south compression of Rocky Mountain foreland structures, in Lowell, James, ed., Rocky Mountains foreland basins and uplifts: Rocky Mountain Association of Geologists Guidebook, p. 9–32.

Harlan, S. S., 1986, Timing of deformation along the leading edge of the Montana Disturbed Belt, northern Crazy Mountains Basin, Montana [M.S. thesis]: Bozeman, Montana, Montana State University, 87 p.

Harlan, S. S., and Lageson, D. R., 1983, Timing of deformation along the leading edge of the fold and thrust belt, Crazy Mountains Basin, Montana: Isochron/West, no. 36, p. 7–8.

Harland, W. B., Cox, A. V., Llewellyn, P. G., Smith, A. G., and Walters, R., 1982, A geologic time scale: Cambridge University Press, 131 p.

Hrouda, F., 1982, Magnetic anisotropy of rocks and its application in geology and geophysics: Geophysical Surveys, v. 5, p. 37–82.

Irving, E., and Irving, G. A., 1982, Apparent polar wander paths Carboniferous through Cenozoic and the assembly of Gondwana: Royal Astronomical Society Geophysical Journal, v. 5, p. 141–188.

Kent, D. V., and Opdyke, N. D., 1986, Multicomponent magnetizations from the Mississippian Mauch Chunk Formation of the central Appalachians and their tectonic implications: Journal of Geophysical Research, v. 90, p. 5371–5383.

Kodama, K. P., 1986, The effect of deformation on the fold test: EOS (American Geophysical Union Transactions), v. 67, no. 16, p. 268.

Larsen, L. H., and Simms, F. E., 1972, Igneous geology of the Crazy Mountains, Montana: A report of work in progress: Montana Geological Society, Annual Field Conference, 21st, Guidebook, p. 135–139.

MacDonald, W. D., 1980, Net tectonic rotation, apparent tectonic rotation, and the structural tilt correction in paleomagnetic studies: Journal of Geophysical Research, v. 85, p. 3659–3669.

Marvin, R. F., and Dobson, S. W., 1979, Radiometric ages: Compilation B, U.S. Geological Survey: Isochron/West, no. 26, p. 3–32.

Marvin, R. F., Witkind, I. J., Keefer, W. R., and Mehnert, H. H., 1973, Radiometric ages of intrusive rocks in the Little Belt Mountains, Montana: Geological Society of America Bulletin, v. 84, p. 1977–1986.

Marvin, R. F., Hearn, B. C., Jr., Mehnert, H. H., Naeser, C. W., Zartman, R. E., Lindsey, D. A., 1980, Late Cretaceous–Paleocene–Eocene igneous activity in north-central Montana: Isochron/West, no. 29, p. 5–25.

McElhinny, M. W., 1964, Statistical significance of the fold test in paleomagnetism: Royal Astronomical Society Geophysical Journal, v. 8, p. 338–340.

McFadden, P. L., and Jones, D. L., 1981, The fold test in paleomagnetism: Royal Astronomical Society Geophysical Journal, v. 67, p. 53–58.

McFadden, P. L., and Lowes, F. J., 1981, The discrimination of mean directions drawn from Fisher distributions: Royal Astronomical Society Geophysical Journal, v. 67, p. 19–33.

McMannis, W. J., 1965, Resume of depositional and structural history of western Montana: American Association of Petroleum Geologists Bulletin, v. 49, p. 1801–1823.

Mehnert, H. H., and Schmidt, R. G., 1971, Dating the Eldorado thrust, in Geological Survey research 1971: U.S. Geological Survey Professional Paper 750-A, p. 37.

Mudge, M. R., 1982, A resume of the structural geology of the northern Disturbed Belt, northwestern Montana, in Geological studies of the Cordilleran thrust belt (Volume 1): Rocky Mountain Association of Geologists, p. 91–122.

Phillips, F. C., 1971, The use of stereographic projections in structural geology: London, England, Edward Arnold Publishers, Ltd., 90 p.

Ramsay, J. G., 1961, The effect of folding upon the orientation of sedimentation structures: Journal of Geology, v. 69, p. 84–100.

Roberts, A. E., 1972, Cretaceous and early Tertiary depositional and tectonic history of the Livingston area, southwestern Montana: U.S. Geological Survey Professional Paper 526-C, 119 p.

Sandberg, S. A., and Butler, R. F., 1985, Paleomagnetic fold test evidence for Laramide age of monoclinal folding, Salt River Canyon, Arizona: Earth and Planetary Science Letters, v. 73, p. 140–146.

Schmidt, C. J., and Garihan, J. M., 1983, Laramide tectonic development of the Rocky Mountain foreland of southwestern Montana, in Lowell, James, ed., Rocky Mountains foreland basins and uplifts: Rocky Mountain Association of Geologists Guidebook, p. 271–294.

Schmidt, C. J., and Hendrix, T. E., 1981, Tectonic controls for thrust belt and Rocky Mountain foreland structures in the northern Tobacco Root Mountains—Jefferson Canyon area, southwestern Montana: Montana Geological Society Southwest Montana Field Conference Guidebook, p. 167–180.

Schmidt, R. G., 1978, Rocks and minerals of the Wolf Creek area, Lewis and Clark and Cascade Counties, Montana: U.S. Geological Survey Bulletin 1441, 91 p.

Simms, F. E., 1965, Hypabyssal alkaline bodies and structure of part of the northwestern Crazy Mountains, Montana: American Mineralogist, v. 50, p. 291–292.

——1966, The igneous petrology, geochemistry, and structural geology of part of the northwestern Crazy Mountains, Montana [Ph.D. dissert.]: Cincinnati, Ohio, University of Cincinnati, 339 p.

Simpson, G. G. 1937, The Fort Union of the Crazy Mountains field, Montana, and its mammalian faunas: U.S. Natural History Museum Bulletin 169, p. 1–287.

Sims, J. D., 1964, A sedimentary and petrographic study of the upper Fort Union Group, northern Crazy Mountains, Montana [M.S. thesis]: Cincinnati, Ohio, University of Cincinnati, 40 p.

——1967, Geology and sedimentology of the Livingston Group, northern Crazy Mountains, Montana [Ph.D. dissert.]: Evanston, Illinois, Northwestern University, 112 p.

Smedes, H. W., and Prostka, H. J., 1972, Stratigraphic framework of the Absaroka Volcanic Supergroup in the Yellowstone National Park region: U.S. Geological Survey Professional Paper 729-C, 33 p.

Stephenson, A., Sadikun, S., and Potter, D. K., 1986, A theoretical and experimental comparison of the anisotropies of magnetic susceptibility and remanence in rocks and minerals: Royal Astronomical Society Geophysical Journal, v. 84, p. 185–200.

Tanner, J. T., 1949, Geology of the Castle Mountain area, Montana [Ph.D. dissert.]: Princeton, New Jersey, Princeton University (unpaginated).

Tysdal, R. G., Marvin, R. F., and Dewitt, E., 1986, Late Cretaceous stratigraphy, deformation, and intrusion in the Madison Range of southwestern Montana: Geological Society of America Bulletin, v. 97, p. 859–868.

van der Pluijm, B. A., 1987, Grain-scale deformation and the fold-test—Evaluation of syn-folding remagnetization: Geophysical Research Letters, v. 14, p. 155–157.

Wilson, C. W., Jr., 1936, Geology of the Nye-Bowler lineament, Stillwater and Carbon Counties, Montana: American Association of Petroleum Geologists Bulletin, v. 20, p. 1161–1188.

Wiltschko, D. J., and Dorr, J. A., Jr., 1983, Timing of deformation in overthrust belt and foreland of Idaho, Wyoming, and Utah: American Association of Petroleum Geologists Bulletin, v. 67, p. 1304–1322.

Wolff, J. E., 1938, Igneous rocks of the Crazy Mountains, Montana: Geological Society of America Bulletin, v. 49, p. 1569–1628.

Woodward, L. A., 1981, Tectonic framework of the Disturbed Belt of west-central Montana: American Association of Petroleum Geologists Bulletin, v. 65, p. 291–302.

Zijderveld, J.D.A., 1967, A. C., Demagnetization of rocks: Analysis of results, in Collinson, D. W., Creer, K. M., and Runcorn, S. K., eds., Methods in paleomagnetism: Amsterdam, the Netherlands, Elsevier, p. 254–286.

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